

CHAPTER3:THERMODYNAMIC PROCESSES ON GASES

Thermodynamic Processes

A process is a path in which the state of the system change and some properties vary from their original values. The prefix **iso-** is often used to designate a process for which a particular property remains constant. The Processes associated with Thermodynamics are:

Isobaric process: is a process during which the pressure P remains constant.

Isochoric (or isometric) process: is a process during which the specific volume remains constant.

isothermal process: is a process during which the temperature T remains constant [5].

Adiabatic process: no heat transfer from or to the fluid.

Isentropic process: no change of entropy of the fluid

Isenthalpic, no change of enthalpy of the fluid.

Application of the First Law of Thermodynamics to Closed or Non-Flow System

The equations of heat and work representing the processes mentioned above are stated below.

The energy equation

The non-flow energy equation for the closed systems derived from the first law of thermodynamics is given as

$$Q + W = U_2 - U_1 \quad 3.1$$

Q = Heat received by the fluid (kJ)

W = work done by the fluid (kJ)

$U_2 - U_1$ = increase in internal energy received by the fluid (kJ)

The relative values of the equation 3.1 depend on the nature of the heat and on the manner in which the work is performed.

Work

If part of the boundary is allowed to move. The work done equation is given as:

$$dW = -PdV \quad 3.2$$

The total work is $W = -\int_1^2 P(dV)$

The negative sign is required because in an expansion dV is positive, whereas dW is negative when the work is done on the surrounding. dV is negative for compression.

Isochoric process equation (volume-constant)

In a constant volume process the working substance is contained in a rigid vessel (**Error! Reference source not found..a**) and the boundaries of the system are immovable and no work can be done on or by the system, other than paddle-wheel work input. It will be assumed that ‘**constant volume**’ implies **zero work** unless stated otherwise.

Error! Reference source not found.b shows the PV diagram of the system and states before and after the heat addition at constant volume. Since the boundary of the system is not movable, then the pressure rises when heat is supplied (**Error! Reference source not found..a**).

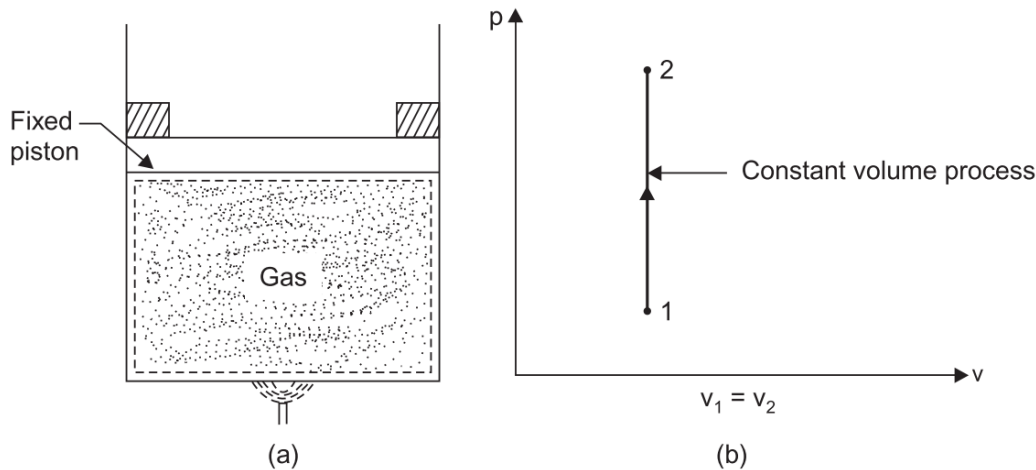


Figure **Error! No text of specified style in document..1** Constant volume process

The energy equation for a constant volume (implies zero work) process is usually written as [6]:

$$Q + W = (u_2 - u_1) \quad 3.3$$

$$Q + 0 = (u_2 - u_1) \quad 3.4$$

$$Q = U_2 - U_1 = mC_v(T_2 - T_1) \quad 3.5$$

Example 3.1

An air in a closed vessel of fixed volume 0.14 m^3 , exerts a pressure of 10 bar at 250°C . If the vessel is cooled so that the pressure falls to 3.5 bar, determine the final temperature and heat transfer.

Note: Air is considered as an example of a perfect gas, and the following values will be used for $C_p = 1.005 \text{ kJ/kgK}$ and $R = 0.287 \text{ kJ/kgK}$ and $C_v = C_p - R$

Solution

Given: $V = 0.14 \text{ m}^3$, $P_1 = 10 \text{ bar} = 10 \times 10^5 \text{ Pa}$, $T_1 = 250 + 273 = 523 \text{ K}$, $P_2 = 3.5 \text{ bar}$, $T_2 = ?$ and $Q = ?$

For constant volume Apply Gay-Lussac's law for gases to find $T_2 = T_1 \frac{P_2}{P_1} =$

$$523 \frac{3.5}{10}$$

$$T_2 = 183K$$

For heat transfer apply $Q = U_2 - U_1 = mC_v(T_2 - T_1) = -228kJ$

Find mass from equation of state $PV=mRT$, where, $m = \frac{P_1V}{RT_1} = \frac{10 \times 10^2 \times 0.14}{0.287 \times 523}$

$$m = 0.933kg$$

Isobaric process equation (constant pressure)

In a constant pressure process, the **boundary must move against** an external resistance as heat is supplied; for instance a gas in Figure Error! No text of specified style in document..2(a) in the cylinder behind a piston can be made to undergo a constant pressure process. Since the piston is pushed through a certain distance by the force exerted by the gas, then the work is done by the gas on its surroundings.

Figure Error! No text of specified style in document..2 shows the PV diagram of the system and states before and after the heat addition at constant pressure.

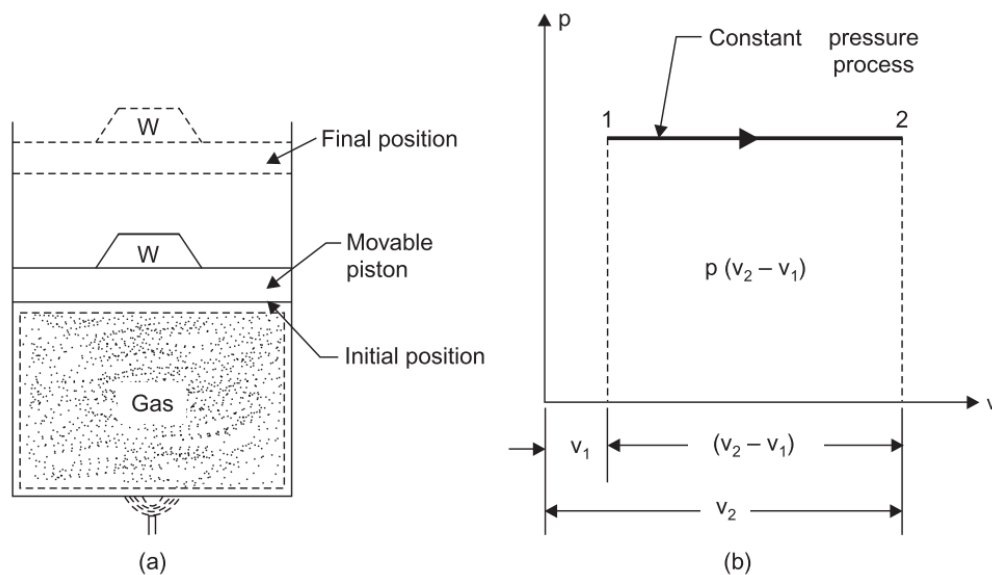


Figure Error! No text of specified style in document..2 constant pressure process

For a constant pressure process undergone by unit mass of any fluid. The non-flow energy equation for a perfect gas is given as:[3].

Note that: $W = -P(V_2 - V_1)$

$$Q + W = u_2 - u_1 \quad 3.6$$

$$dQ - pdV = du \quad 3.7$$

$$dQ = pdV + du$$

$$dQ = d(PV + U) = dH \quad 3.8$$

$$Q = h_2 - h_1 = C_P(T_2 - T_1) \quad 3.9$$

For mass, m, of working substance

$$Q = H_2 - H_1 = mC_P(T_2 - T_1)$$

Example 3.2

A mass of 0.2 kg of air, initially at a temperature of 165°C, expands reversibly at a constant pressure of 7 bar until the volume is doubled. Find the final temperature, work and heat transfer.

Solution

Given: $m = 0.2\text{kg}$, $T_1 = 165 + 273 = 438\text{K}$, $P_1 = 7\text{bar} = 7 \times 10^2 \text{kPa}$, $V_1 = V_2$
 $T_2 = ?$, $W = ?$ and $Q = ?$

For constant pressure Apply Charle's law for gases to find $T_2 = T_1 \frac{V_2}{V_1} = 876\text{K}$

In order to find V_2 & V_1 apply $PV = mRT$, $V_1 = 0.0359\text{m}^3$ and $V_2 = 0.0718\text{m}^3$

For work done apply

$$W = -P(V_2 - V_1) = -25$$

For Heat transfer apply

$$Q = U_2 - U_1 = mC_v(T_2 - T_1) = 88\text{kJ}$$

Isothermal process equation (Temperature-constant)

A process at a constant temperature is called an isothermal process. When a working substance in a cylinder behind a piston expands from a high pressure to a low pressure there is a tendency for the temperature to fall.

In an isothermal expansion, heat must be added continuously in order to keep the temperature at the initial value. Similarly in an isothermal compression heat must be removed from the working substance continuously during the process. Figure Error! No text of specified style in document..3 shows the system and states before and after the heat addition at a constant temperature.

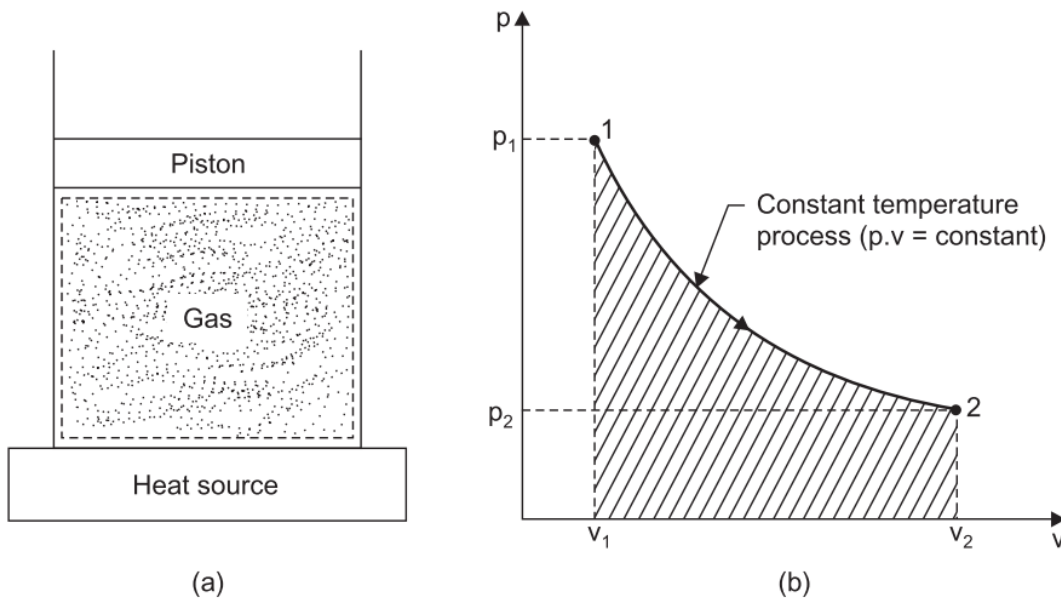


Figure Error! No text of specified style in document..3 Isothermal process.

The non-flow energy equation is given as

$$dQ - Pdv = du \quad 10$$

$$\text{Or } Q - \int_1^2 P(dV) = u_2 - u_1$$

$$W = (u_2 - u_1) - Q \quad 11$$

Adiabatic process equation

An **adiabatic process** is one in which no heat is transferred to or from the fluid during the process. Such a process can be reversible or irreversible. The reversible adiabatic non-flow process

$$Q + W = (u_2 - u_1)$$

$$0 + W = (u_2 - u_1)$$

For adiabatic non-flow processes, the energy equation reduces to

$$W = u_2 - u_1 \qquad 12$$

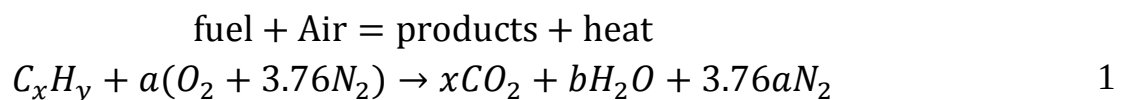
CHAPTER 4: FUEL AND COMBUSTION PROCESSES

Fuel and combustion

Any material that can be burned to release thermal energy is called a **fuel**. Most familiar fuels consist primarily of hydrogen and carbon. They are called hydrocarbon fuels and are denoted by the general formula C_xH_y . Hydrocarbon fuels exist in all phases (solid, liquid and gas), some examples being coal, gasoline, and natural gas.

A chemical reaction during which a fuel is oxidized and a large quantity of energy is released is called **combustion**. The oxidizer most often used in combustion processes is air: Pure oxygen, O_2 is used as an oxidizer only in some specialized applications, such as cutting and welding, where air cannot be used. Thus, Combustion is the self-sustaining rapid exothermic chemical reaction between the elemental C, H, and S of fuel with O_2 from air due to their affinity. The products of combustion are CO_2 , H_2O , and SO_2 along with heat. C and H are beneficial constituents of fuel, producing heat, whereas S is the least heat generating and most harmful, as the SO_2 produced causes low-temperature corrosion. To make any combustion self-sustaining, the resultant temperature from the exothermic reaction should exceed the ignition temperature of the fuel.

Combustion is the reaction of a fuel substance with air to form combustion products. The combustion process is always exothermic and it liberates heat [7]. The combustion reaction may be written as



The main combustion products of burning of fuels are carbon dioxide and water.

Applications of Combustion

The area of application of combustion include: Power Plants, Chemical Industries, Domestic Burner (in cooking and bakery), and Automobiles, etc.

Types of fuels

There are three types of fuels; solid fuels, liquid fuels and gaseous fuels, all of which have different advantages.

Gaseous fuels can be handled most easily, in terms of both purification and distribution, with low requirements for system maintenance, they are generally burn in a very clean way. **Gaseous fuels include;** Natural Gas, Liquefied Petroleum Gas (LPG), Propane, Butane, Hydrogen etc. They are used in both domestic and industrial applications

Liquid fuels have a high energy density and are hence preferably used in the transportation industries, where they are burnt in engines (internal combustion engines, gas turbines). **Liquid fuels include;** kerosene, petrol (Gasoline), diesel, Fuel Oil, alcohol, ethanol, butanol, methanol etc. They are used in both domestic and industrial applications.

Solid fuels can be stored in large quantities [8]. Examples of **Solid fuels** include, Wood, coal, peat, coke, charcoal etc. Solid fuels can be fossil (e.g. coal) or renewable (e.g. wood or bagasse). They are used in both domestic and industrial applications.

Various applications of different types of fuels

Substances that produce heat and light energy on burning are called fuels. Generally, fuels are used for **cooking, heating, in transportation industries, and for the generation of electricity**. Fuel selection depends on fuel characteristics, availability and costs.

Calorific or Heating Values of Fuel, Dulong's formula for calorific value

Calorific value is the amount of heat energy present in fuel and which is determined by the complete combustion of specified quantity at constant pressure and in normal conditions. The unit of calorific value is kilojoule per kilogram i.e. kJ/kg for solid and liquid fuels and kJ/m³ for gaseous fuels.

If a fuel contains hydrogen water will be formed as one of the products of combustion. If this water is condensed, a large amount of heat will be released than if the water exists in the vapor phase. For this reason two heating values are defined; the **higher or gross** heating value and the **lower or net heating** value.

The higher heating value (HHV), is obtained when the water formed by combustion is completely condensed. **The lower heating value (LHV)**, is obtained when the water formed by combustion exists completely in the vapor phase. Thus:

$$(HHV)_p = (LHV)_p + mh_{fg}$$

$$(HHV)_v = (LHV)_v + m(u_g - u_f)$$

Where m = Mass of water formed by combustion, h_{fg} = Enthalpy of vaporization of water, kJ/kg, u_g = Specific internal energy of vapor, kJ/kg, and u_f = Specific internal energy of liquid, kJ/kg.

In almost all practical cases, the water vapor in the products is vapor, the lower value is the one which usually applies.

Complete, incomplete and stoichiometric combustion

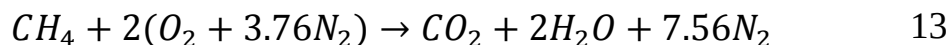
A combustion process is **complete** if all the carbon in the fuel burns to CO₂, all the hydrogen burns to H₂O, and all the sulfur (if any) burns to SO₂. That is, all the combustible components of the fuel are burned to completion during a complete combustion process. In complete combustion, the product mixture consists of only CO₂, H₂O, and N₂.

The combustion process is **incomplete** if the combustion products contain any unburned fuel or components such as C, H₂, CO, or OH.

Insufficient oxygen is an obvious reason for incomplete combustion, but it is not the only one. Incomplete combustion occurs even when more oxygen is present in the combustion chamber than is needed for complete combustion. This may be attributed to insufficient mixing in the combustion chamber during the limited time that the fuel and the oxygen are in contact. Another cause of incomplete combustion is **dissociation**, which becomes important at high temperatures.

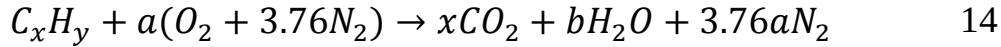
Oxygen has a much greater tendency to combine with hydrogen than it does with carbon. Therefore, the hydrogen in the fuel normally burns to completion, forming H₂O, even when there is less oxygen than needed for complete combustion. Some of the carbon, however, ends up as CO or just as plain C particles (soot) in the products.

The minimum amount of air needed for the complete combustion of a fuel is called **stoichiometric** or **theoretical air**. Thus, when a fuel is completely burned with theoretical air, no uncombined oxygen is present in the product gases. The theoretical air is also referred to as the *chemically correct amount of air*, or *100 percent theoretical air*. A combustion process with less than the theoretical air is bound to be incomplete. The ideal combustion process during which a fuel is burned completely with theoretical air is called the **stoichiometric** or **theoretical combustion** of that fuel. For example, the theoretical combustion equation of methane is given as:



Stoichiometric relation

For a hydrocarbon fuel (C_x H_y) in Air. The stoichiometric combustion relation can be expressed as



Where, Air = 21% oxygen and 79% Nitrogen, thus one mole of O_2 combine with (79/21) mole of N_2 which gives $O_2 + 3.76N_2$,

$$a = x + \frac{y}{4}, \text{ and } b = \frac{y}{2}.$$

Air fuel ratio, rich mixture and lean mixture

Air–fuel ratio AF. It is usually expressed on a mass basis and is defined as the ratio of the mass of air to the mass of fuel for a combustion process. The stoichiometric **air-fuel ratio** can be calculated as:

$$\left(\frac{A}{F}\right)_{Stoic} = \frac{\dot{m}_{air}}{\dot{m}_{fuel}} = \frac{4.76aMW_{air}}{1 MW_{fuel}}$$

Where MW_{air} and MW_{fuel} are the molecular weights of the air (29kg/kmol) and fuel (depending on the chemical composition of the fuel), respectively. The mass m of a substance is related to the number of moles N through the relation $m = NM$, where M is the molar mass.

Lean; excess of air (more air than needed by fuel). If more than a stoichiometric quantity of air is supplied, the mixture is said to be fuel lean. It is also called a weak mixture is one that has an excess of air.

Rich mixture or fuel-rich- A rich mixture is one that has a deficiency of air. Or excess fuel (less air than needed by fuel).

Table 4.1 Symbols and Molecular weights of some important elements, compounds and gases

Elements/Compounds/Gases	Molecule		Atom	
	Symbol	Molecular Weight	Symbol	Molecular Weight
Hydrogen	H ₂	2	H	1

Oxygen	O ₂	32	O	16
Carbon	C	12	C	12
Methane	CH ₄	(12x1)+(1x4)=16		
Ethane	C ₂ H ₄	(12x2)+(1x4)=30		
Propane	C ₃ H ₈	(12x3)+(1x8)=44		
Pentane	C ₅ H ₁₂	(12x5)+(1x12)=72		
Hexane	C ₆ H ₁₄	(12x6)+(1x14)=86		
Air		29		

A combustion process is **complete** if all the carbon in the fuel burns to CO₂, all the hydrogen burns to H₂O, and all the sulfur (if any) burns to SO₂. That is, all the combustible components of the fuel are burned to completion during a complete combustion process. In complete combustion, the product mixture consists of only CO₂, H₂O, and N₂.

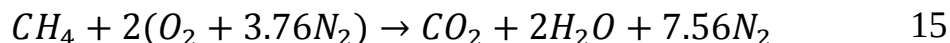
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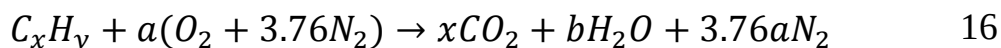
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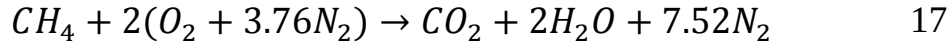
Where, Air = 21% oxygen and 79% Nitrogen, thus one mole of O_2 combine with (79/21) mole of N_2 which gives $O_2 + 3.76N_2$, $a = x + \frac{y}{4}$, and $b = \frac{y}{2}$.

Calculate the theoretical/stoichiometric quantity of air required

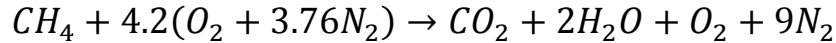
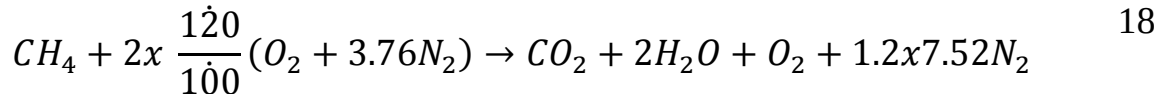
The minimum amount of air that supplies sufficient oxygen for the complete combustion of all the carbon, hydrogen, and any other elements in the fuel that may oxidize is called the “**theoretical air**”. When complete combustion is achieved with theoretical air, the products contain no oxygen.

In practice, it is not possible to achieve complete combustion. Thus, the amount of air to be supplied must be greater than the theoretical amount. For example 120 percent theoretical air means that the air actually supplied is 1.2 times the theoretical air.

The complete combustion of methane with the minimum amount of theoretical air and 120 percent theoretical air respectively is written as:



With 120 percent theoretical air



The amount of air actually supplied may also be expressed in terms of percent excess air. **The excess air** is the amount of air supplied over and above the theoretical air. Thus 120 percent theoretical air is equivalent to 20 percent excess air.

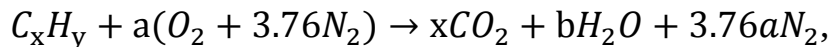
Examples

1. Determine the combustion equations and molar masses of the following fuels: i) Hexane C_6H_{14} , ii) Propane C_3H_8 and iii) Pentane C_5H_{12} .

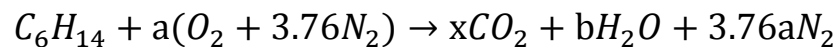
Solution

Remember the definition of combustion:

Combustion is the reaction of a fuel substance with air to form combustion products. The general formula for combustion equation **is given as:**



Thus, for hexane is

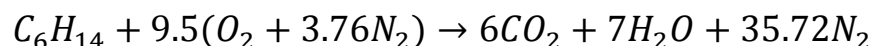


$$x = 6$$

$$a = x + \frac{y}{4} = 6 + \frac{14}{4} = 6 + 3.5 = 9.5$$

$$b = \frac{y}{2} = \frac{14}{2} = 7$$

The equation becomes



Or by balancing the moles of the reactants and products

The unknowns **a**, **x** and **b** can also be determined by balancing mole number of each element in the reactants be equal to that in the products:

$$\text{Carbon, C:} \quad 6 = x \rightarrow x = 6$$

$$\text{Hydrogen, H:} \quad 14 = 2b \rightarrow b = 7$$

$$\begin{aligned} \text{Oxygen, O}_2 \quad 2a &= 2x + b = (2 \times 6) + 7 \\ 2a &= 19 \rightarrow a = 9.5 \end{aligned}$$

2. Octane (C_8H_{18}) is burned completely with atmospheric air, Determine the:
(i) Combustion equation; (ii) mole number of each gas in the products, and
iii) stoichiometric air/fuel ratio.

Solution

$$(i) \quad C_8H_{18} + a(O_2 + 3.76N_2) \rightarrow xCO_2 + bH_2O + 3.76aN_2$$

(ii) mole number of each gas in the products

The unknowns **a**, **x** and **b** can be determined by equating the mole number of each element in the reactants and the products:

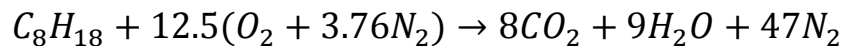
$$\text{Carbon, C:} \quad 8 = x \rightarrow x = 8$$

$$\text{Hydrogen, H:} \quad 18 = 2b \rightarrow b = 9$$

$$\begin{aligned} \text{Oxygen, O}_2 \quad 2a &= 2x + b = (2 \times 8) + 9 \\ 2a &= 25 \rightarrow a = 12.5 \end{aligned}$$

$$N_2 = 3.76 \times 12.5 = 47$$

Thus, mole number of each gas in the products are: $x = 8$, $b = 9$ and $N_2 = 47$



iii) stoichiometric air/fuel ratio:

$$\left(\frac{A}{F}\right)_{Stoic} = \frac{\dot{m}_{air}}{\dot{m}_{fuel}(C_8H_{18})} = \frac{4.76 \times 12.5 \times 29}{(12 \times 8) + (1 \times 18)} = \frac{1725.5}{114} = 15.14$$